

A New Compressible Multiphase Flow Solver for Complex System Modeling Platform (CSMP++)

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Governing Equations

For simplicity, but without loss of generality, no dissolution and no capillary effects are considered here. In this case, Darcy's formulation of multiphase flow in porous media leads to the mass conservation law

$$\frac{\partial}{\partial t}(\phi \rho_\alpha S_\alpha) - \nabla \cdot (\rho_\alpha \lambda_\alpha \mathbf{K} \cdot (\nabla p - \rho_\alpha g \nabla z)) = \rho_\alpha q_\alpha \quad \forall \alpha \in \{1, \dots, \gamma\} \quad (1)$$

of phase α , where ϕ denotes the porosity, $\mathbf{K}$ the positive definite permeability tensor, ρ_α the density, S_α the saturation, g the gravitational acceleration, and q_α source terms. Moreover, $\lambda_\alpha = k_{r\alpha}/\mu_\alpha$ is the phase mobility with the relative permeability $k_{r\alpha}$ and the phase viscosity μ_α . Porosity, density and viscosity are given as algebraic functions of the pressure p and $k_{r\alpha}$ is a function of the saturations. Phase saturations are subject to the constraint

$$\sum_{\alpha=1}^{\gamma} S_\alpha = 1. \quad (2)$$

Note that Eqs. (1) and (2) form a system for $\gamma + 1$ unknowns $(S_1, \dots, S_\alpha, p)$ [1].

Implicit Euler time integration leads to

$$\frac{\phi^{n+1} \rho_\alpha^{n+1} S_\alpha^{n+1} - \phi^n \rho_\alpha^n S_\alpha^n}{\Delta t} - \nabla \cdot (\rho_\alpha^{n+1} \lambda_\alpha^{n+1} \mathbf{K} \cdot (\nabla p^{n+1} - \rho_\alpha^{n+1} g \nabla z)) = \rho_\alpha^{n+1} q_\alpha, \quad (3)$$

which is now divided by the density ρ_α^{n+1} in order to obtain a decoupled pressure equation. The super index n denotes the time step level. Taking the sum over all phases α results in

$$\frac{\phi^{n+1}}{\Delta t} - \frac{\phi^n}{\Delta t} \sum_{\alpha=1}^{\gamma} B_\alpha^{n+1} \rho_\alpha^n S_\alpha^n - \sum_{\alpha=1}^{\gamma} B_\alpha^{n+1} \nabla \cdot (\rho_\alpha^{n+1} \lambda_\alpha^{n+1} \mathbf{K} \cdot (\nabla p^{n+1} - \rho_\alpha^{n+1} g \nabla z)) = q, \quad (4)$$

where $q = \sum_{\alpha=1}^{\gamma} q_{\alpha}$ denotes the total volumetric source term and $B_{\alpha} = 1/\rho_{\alpha}$ is the volume formation factor. The pressure dependent parameters are now linearized using $\kappa^{n+1} \approx \kappa^{\nu_p+1} \approx \kappa^{\nu_p} + \frac{\partial \kappa}{\partial p} |^{\nu_p} (p^{\nu_p+1} - p^{\nu_p})$ and the saturation dependent mobilities are calculated based on the most updated saturation solution ν_s resulting in a sequentially fully implicit scheme. Therefore, the iterative linear pressure equation reads

$$\frac{C}{\Delta t} (p^{\nu_p+1} - p^{\nu_p}) - \sum_{\alpha=1}^{\gamma} B_{\alpha}^{\nu_p} \nabla \cdot (\rho_{\alpha}^{\nu_p} \lambda_{\alpha}^{\nu_s} \mathbf{K} \cdot \nabla p^{\nu_p+1}) = -\frac{\phi^{\nu_p}}{\Delta t} + \frac{\phi^n}{\Delta t} \sum_{\alpha=1}^{\gamma} B_{\alpha}^{\nu_p} \rho_{\alpha}^n S_{\alpha}^n + \quad (5)$$

$$q - \sum_{\alpha=1}^{\gamma} B_{\alpha}^{\nu_p} \nabla \cdot (\rho_{\alpha}^{\nu_p} \lambda_{\alpha}^{\nu_s} \mathbf{K} g \cdot \nabla z),$$

which converges to Eq. (4) as pressure and saturation iteration indices tend to infinity. In this equation

$$C = \frac{\partial \phi}{\partial p} |^{\nu_p} - \phi^n \sum_{\alpha=1}^{\gamma} \frac{\partial B_{\alpha}}{\partial p} |^{\nu_p} \rho_{\alpha}^n S_{\alpha}^n \quad (6)$$

is the compressibility coefficient. Note that the coefficients B_{α} and ρ_{α} in the convective and gravitational terms lag one iteration behind. This discretized formulation was already used in previous studies to solve the fine-scale problem [2, 3].

Here, the convective term is consistently modified as

$$\sum_{\alpha=1}^{\gamma} B_{\alpha}^{\nu_p} \nabla \cdot (\rho_{\alpha}^{\nu_p} \lambda_{\alpha}^{\nu_s} \mathbf{K} \cdot \nabla p^{\nu_p+1}) \approx \quad (7)$$

$$\nabla \cdot (\lambda_t^{\nu_s} \mathbf{K} \cdot \nabla p^{\nu_p+1}) - \nabla \cdot (\lambda_t^{\nu_s} \mathbf{K} \cdot \nabla p^{\nu_p}) + \sum_{\alpha=1}^{\gamma} B_{\alpha}^{\nu_p} \nabla \cdot (\rho_{\alpha}^{\nu_p} \lambda_{\alpha}^{\nu_s} \mathbf{K} \cdot \nabla p^{\nu_p}),$$

where $\lambda_t^{\nu_s} = \sum_{\alpha=1}^{\gamma} \lambda_{\alpha}^{\nu_s}$ is the total mobility [4, 5]. Note that $\nu_p = \nu_p + 1 = n + 1$, if converged. It is worth mentioning that for black-oil models, which are more realistic to describe multiphase flow in oil reservoirs [1], one can still use this expression. Additional effects are e.g. capillary pressure differences and mass exchange between phases, which are treated on the right hand side, i.e. the implicit convection operator has the same structure as in Eq. (7) [3]. Finally, our pressure equation reads

$$\frac{C}{\Delta t} (p^{\nu_p+1} - p^{\nu_p}) - \nabla \cdot (\lambda_t^{\nu_s} \mathbf{K} \cdot \nabla p^{\nu_p+1}) = RHS^{\nu_p}, \quad (8)$$

where

$$\begin{aligned}
RHS^{\nu_p} = & -\frac{\phi^{\nu_p}}{\Delta t} + \frac{\phi^n}{\Delta t} \sum_{\alpha} B_{\alpha}^{\nu_p} \rho_{\alpha}^n S_{\alpha}^n + q - \nabla \cdot (\lambda_t^{\nu_s} \mathbf{K} \cdot \nabla p^{\nu_p}) \\
& \sum_{\alpha}^{\gamma} B_{\alpha}^{\nu_p} \nabla \cdot (\rho_{\alpha}^{\nu_p} \lambda_t^{\nu_s} f_{\alpha}^{\nu_s} \mathbf{K} \cdot (\nabla p^{\nu_p} - \rho_{\alpha}^{\nu_p} g \cdot \nabla z)),
\end{aligned} \tag{9}$$

which is solved iteratively until convergence occurs, $\nu_p = \nu_p + 1 = n + 1$. Here, $\lambda_{\alpha}^{\nu_s}$ is replaced with $f_{\alpha}^{\nu_s} \lambda_t^{\nu_s}$, where $\lambda_t^{\nu_s}$ is computed at the cell interfaces using a harmonic averaging scheme between neighboring cells. Within each pressure iteration the saturation equations are iteratively solved in an inner-loop denoted by super index ν_s , i.e.

$$\begin{aligned}
& \frac{\phi^{\nu_p+1} \rho_{\alpha}^{\nu_p+1} S_{\alpha}^{\nu_s+1} - \phi^n \rho_{\alpha}^n S_{\alpha}^n}{\Delta t} \\
& \nabla \cdot \left(\rho_{\alpha}^{\nu_p+1} \left(f_{\alpha}^{\nu_s} + \frac{\partial f_{\alpha}}{\partial S_{\alpha}} \Big|_{\nu_s} (S_{\alpha}^{\nu_s+1} - S_{\alpha}^{\nu_s}) \right) \mathbf{u}_t^{\nu_s} \right) = \rho_{\alpha}^{\nu_p+1} q_{\alpha},
\end{aligned} \tag{10}$$

where $\mathbf{u}_t^{\nu_s} = \lambda_t^{\nu_s} \mathbf{K} \cdot (\nabla p^{\nu_p+1} - \rho_{\alpha}^{\nu_p+1} g \nabla z)$. This iterative loop is repeated ($\nu_s \leftarrow \nu_s + 1$) till convergence occurs. Then the new saturation dependent parameters are inserted into the pressure equation and the procedure is repeated till the decoupled-pressure-saturation system of equation converges. The coefficients f_{α} are computed using a second order upwind scheme with the convective speed

$$\frac{\partial}{\partial(\phi \rho_{\alpha} S_{\alpha})} (\rho_{\alpha} \lambda_{\alpha} \mathbf{K} \cdot (\nabla p - \rho_{\alpha} g \nabla z)). \tag{11}$$

Since the pressure dependent coefficients are constant in the saturation update step, Eq. (11) is simplified to

$$\frac{1}{\phi} \mathbf{K} \cdot (\nabla p - \rho_{\alpha} g \nabla z) \frac{\partial \lambda_{\alpha}}{\partial S_{\alpha}}, \tag{12}$$

which has the same sign as the phase velocity for $\frac{\partial \lambda_{\alpha}}{\partial S_{\alpha}} > 0$ and positive definite permeability fields.

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